

Math 210B: Homework 1

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Problem 1 (Algebra Comp Fall 2021 Rings). Let R be a commutative ring with unity. An element $a \in R$ is called **nilpotent** if $a^n = 0$ for some $n \in \mathbb{N}$ that can depend on a . Let $N \subseteq R$ be the subset of all nilpotent elements of R .

(a) Prove that N is an ideal of R .

Proof. First, we show that N is an additive subgroup of R . Note that $N \neq \emptyset$ because $0^n = 0$ for any $n \in \mathbb{N}$. Now, let $x, y \in N$ be arbitrary. Our goal is to show that $x + (-y) \in N$ using the Binomial Theorem. Since $x \in N$ and $y \in N$, it follows that $x^n = 0$ and $y^m = 0$ for some $n, m \in \mathbb{N}$, respectively. Then we have

$$\begin{aligned}(x + (-y))^{nm} &= (x - y)^{nm} = \sum_{k=0}^{nm} \binom{nm}{k} x^{n-k} y^k \\&= \binom{nm}{0} x^{nm} y^0 + \binom{nm}{1} x^{nm-1} y^1 + \binom{nm}{2} x^{nm-2} y^2 + \cdots \\&\quad + \binom{nm}{nm-1} x^1 y^{nm-1} + \binom{nm}{nm} x^0 y^{nm} \\&= \binom{nm}{0} (x^n)^m y^0 + \binom{nm}{1} (x^n)^m \cdot x^{-1} + \binom{nm}{2} x^{nm} \cdot x^{-2} \cdot y^2 \\&\quad + \cdots + \binom{nm}{nm-1} x^1 \cdot (y^m)^n \cdot y^{-1} + \binom{nm}{nm} x^0 (y^m)^n \\&= 0 + 0 + 0 + \cdots + 0 + 0 \\&= 0.\end{aligned}$$

This tells us that $(x + (-y))^{nm} = 0$ and so $x + (-y) \in N$. Thus, we conclude that N is an additive subgroup of R .

Now, we prove that N is closed under multiplication. Let $r \in R$. Our goal is to show that for all $r \in R$,

- (i) $rN \subseteq N$,
- (ii) $Nr \subseteq N$.

Let $r \in R$ be arbitrary.

(i) Let $y \in rN$. Then $y = rx$ for some $x \in N$. Since $x \in N$, $x^n = 0$ for some $n \in \mathbb{N}$. Since R is a commutative ring, it can be shown via induction that for any $k \in \mathbb{N}$, $(rs)^k = r^k s^k$ for all $r, s \in R$. In particular, we get that

$$y^n = (rx)^n = r^n x^n = r^n \cdot 0_R = 0_R.$$

Thus, it follows that $y \in N$.

(ii) Similarly, let $y \in Nr$. Then $y = zr$ for some $z \in N$. Since $z \in N$, there exists an $m \in \mathbb{N}$ such that $z^m = 0$. Then

$$y^m = (zr)^m = z^m r^m = 0_R \cdot r^m = 0_R.$$

Thus, it follows that $y \in N$. Since N is closed under left and right multiplication, it follows that N must be an ideal of R . ■

- (b) Prove that the quotient ring R/N contains no nonzero nilpotent elements.

Proof. Suppose for contradiction that R/N contains nonzero nilpotent elements. That is, there exists at least one $r + N \in R/N$ where $r + N \neq 0_R + N$ such that $(r + N)^n = 0_R + N$ for some $n \geq 1$. One can prove via induction that

$$(r + N)^n = r^n + N.$$

Since $(r + N)^n = 0_R + N$, $r^n \in N$ and so for some $m \in \mathbb{N}$, $(r^n)^m = 0_R$. Furthermore, we see that

$$(r^n)^m + N = (r^n + N)^m = (0_R + N)^m = 0_R + N.$$

But this tells us that $(r^n + N)^m = 0_R + N$ which is a zero nilpotent element of R/N , a contradiction. ■

Problem 2 (Section 7.2 Problem 17). Let R and S be nonzero rings with identity and denote their respective identities by 1_R and 1_S . Let $\varphi : R \rightarrow S$ be a nonzero homomorphism of rings.

- (a) Prove that if $\varphi(1_R) \neq 1_S$, then $\varphi(1_R)$ is a zero divisor in S . Deduce that if S is an integral domain then every ring homomorphism from R to S sends the identity of R to the identity of S .

Proof. Our goal is to show that $\varphi(1_R)$ is a zero divisor in S . That is, we need to find a $s \in S$ that is nonzero such that either $\varphi(1_R) \cdot s = 0_S$ or $s \cdot \varphi(1_R) = 0_S$. Since $\varphi(1_R) \neq 1_S$, define $s = \varphi(1_R) - 1_S \neq 0_S$. Then observe that

$$\begin{aligned} \varphi(1_R)[\varphi(1_R) - 1_S] &= [\varphi(1_R)]^2 - \varphi(1_R) \\ &= \varphi(1_R^2) - \varphi(1_R) \\ &= \varphi(1_R) - \varphi(1_R) \\ &= 0_S. \end{aligned}$$

Hence, we conclude that $\varphi(1_R)$ is a zero divisor of S .

We proceed to prove the next result via contrapositive. That is, we want to show that S is NOT an integral domain. To this end, we need to show that there exists an element s in S such that s is a zero divisor. From our assumption, there exists a ring homomorphism $\psi : R \rightarrow S$ such that $\psi(1_R) \neq 1_S$. Then by our previous result, $\psi(1_R)$ is a zero divisor in S ; that is, $s = \psi(1_R) \neq 0$. Hence, S is not an integral domain. ■

- (b) Prove that if $\varphi(1_R) = 1_S$, then $\varphi(u)$ is a unit in S and that $\varphi(u^{-1}) = \varphi(u)^{-1}$ for each unit u of R .

Proof. Let u be a unit in R . By definition, there exists a $v \in R$ such that $uv = vu = 1_R$. Since $\varphi : R \rightarrow S$ is a homomorphism, it follows that

$$\begin{aligned} \varphi(uv) &= \varphi(1_R) = 1_S \\ \implies \varphi(u)\varphi(v) &= 1_S \\ \implies \varphi(u) &= [\varphi(v)]^{-1} \end{aligned}$$

and

$$\begin{aligned} \varphi(vu) &= \varphi(1_R) = 1_S \\ \implies \varphi(v)\varphi(u) &= 1_S \\ \implies \varphi(u) &= [\varphi(v)]^{-1}. \end{aligned}$$

This tells us that we must have $[\varphi(v)]^{-1} = [\varphi(u)]^{-1}$. ■

Problem 3 (Problem 18 Section 7.3). (a) If I and J are ideals of R , prove that $I \cap J$ is also an ideal of R .

Proof. First, we show that $I \cap J$ is an additive subgroup of R . Since I and J are ideals of R , it follows that I and J are also additive subgroups. From 210A, we know that $I \cap J$ must also be an additive subgroup. Our goal is to show that for all $r \in R$

$$(i) \quad r(I \cap J) \subseteq I \cap J$$

$$(ii) \quad (I \cap J)r \subseteq I \cap J$$

Let $r \in R$ be arbitrary.

(i) Let $y \in r(I \cap J)$. Then for some $x \in I \cap J$, $y = rx$. Then $x \in I$ and $x \in J$ which implies that

$$y = rx \in rI \text{ and } y = rx \in rJ.$$

Since I and J are ideals, it follows that $rI \subseteq I$ and $rJ \subseteq J$. Hence, $y \in I$ and $y \in J$, implying that $y \in I \cap J$. This tells us that $r(I \cap J) \subseteq I \cap J$.

(ii) Similarly, let $y \in (I \cap J)r$. Then for some $z \in I \cap J$, $y = zr$. Since $z \in I \cap J$, it follows that $z \in I$ and $z \in J$. This tells us that $y \in Ir$ and $y \in Jr$. Since I and J are ideals, we have that $Ir \subseteq I$ and $Jr \subseteq J$. Thus, we have

$$y \in Ir \cap Jr \subseteq I \cap J \implies y \in I \cap J.$$

Therefore, $(I \cap J)r \subseteq I \cap J$.

Thus, (i) and (ii) imply that $I \cap J$ must also be an ideal of R . ■

(b) Prove that the intersection of an arbitrary nonempty collection of ideals is again an ideal.

Proof. Let $\{I_\alpha\}_{\alpha \in \Lambda}$ be an arbitrary collection of ideals. Denote the arbitrary intersection as

$$I = \bigcap_{\alpha \in \Lambda} I_\alpha.$$

First, we show that I is an additive group. Since each I_α is an additive subgroup, it follows that each $I_\alpha \neq \emptyset$ and for all $x, y \in I_\alpha$, $x + (-y) \in I_\alpha$ for all $\alpha \in \Lambda$. Hence, $I \neq \emptyset$ and that $x + (-y) \in I_\alpha$, implying that I is an additive group. We need to show that for all $r \in R$,

$$(i) \quad rI \subseteq I,$$

$$(ii) \quad Ir \subseteq I.$$

(i) Let $y \in rI$. Then for some $x \in I$, $y = rx$. Since $I = \bigcap_{\alpha \in \Lambda} I_\alpha$, it follows that $x \in I_\alpha$ for all $\alpha \in \Lambda$. Thus, $y \in rI_\alpha$ for all $\alpha \in \Lambda$. Since each I_α is an ideal, we have $rI_\alpha \subseteq I_\alpha$ for all $\alpha \in \Lambda$. Then it follows that $y \in I_\alpha$ for all $\alpha \in \Lambda$. Thus, $y \in I$ and so $rI \subseteq I$.

(ii) Let $y \in Ir$. Then for some $z \in I$, $y = zr$. Since $I = \bigcap_{\alpha \in \Lambda} I_\alpha$, it follows that $z \in I_\alpha$ for all $\alpha \in \Lambda$. Thus, $y \in I_\alpha r$ for all $\alpha \in \Lambda$. Since each I_α is an ideal, we have $I_\alpha r \subseteq I_\alpha$ for all $\alpha \in \Lambda$. Then it follows that $y \in I_\alpha$ for all $\alpha \in \Lambda$. Hence, $y \in I$ and so $Ir \subseteq I$.

Now, (i) and (ii) implies that I must be an ideal of R . Thus, the arbitrary intersection of ideals must also be an ideal. ■

Problem 4. Let $\varphi : R \rightarrow S$ be a ring homomorphism.

(a) Prove that if J is an ideal of S , then $\varphi^{-1}(J)$ is an ideal of R . Apply this to the special case when R is a subring of S and φ is the inclusion homomorphism to deduce that if J is an ideal of S , then $J \cap R$ is an ideal of R .

Proof. Note that since J is an additive subgroup of S , it follows from one of our homeworks from last semester that $\varphi^{-1}(J)$ is an additive subgroup of R . Now, we need to show that $\varphi^{-1}(J)$ is closed under multiplication. To this end, we need to show that for all $r \in R$,

$$(i) \quad r\varphi^{-1}(J) \subseteq \varphi^{-1}(J),$$

$$(ii) \quad \varphi^{-1}(J)r \subseteq \varphi^{-1}(J)$$

(i) Let $y \in r\varphi^{-1}(J)$. Then for some $x \in \varphi^{-1}(J)$, $y = rx$. Since $x \in \varphi^{-1}(J)$, we have $\varphi(x) \in J$. Thus,

$$\varphi(y) = \varphi(rx) = \varphi(r)\varphi(x) \in \varphi(r)J.$$

Since J is an ideal, we have $\varphi(r)J \subseteq J$. Hence, $\varphi(y) \in J$ implying that $y \in \varphi^{-1}(J)$. Thus, $r\varphi^{-1}(J) \subseteq \varphi^{-1}(J)$.

(ii) Similarly, let $y \in \varphi^{-1}(J)r$. Then for some $v \in \varphi^{-1}(J)$, we have $y = vr$. Since $v \in \varphi^{-1}(J)$, it follows that $\varphi(v) \in J$. Thus,

$$\varphi(y) = \varphi(vr) = \varphi(v)\varphi(r) \in J\varphi(r).$$

Since J is an ideal, it follows that $J\varphi(r) \subseteq J$. Hence, $\varphi(y) \in J$ and so $y \in \varphi^{-1}(J)$. This tells us that $\varphi^{-1}(J)r \subseteq \varphi^{-1}(J)$.

Suppose φ is the inclusion map. That is, $\varphi(x) = x$ for all $x \in R$ and $R \subseteq S$. We need to show that for all $r \in R$, we have $r(J \cap R) \subseteq J \cap R$ and $(J \cap R)r \subseteq J \cap R$.

(i) Let $y \in r(J \cap R)$ be arbitrary. By definition, we have that $y = rx$ for some $x \in J \cap R$. Thus, $x \in J$ and $x \in R$. Using φ , it follows that $x = \varphi(x) \in J$, and so $x \in \varphi^{-1}(J)$. Since $\varphi^{-1}(J)$ is an ideal, it follows that $r\varphi^{-1}(J) \subseteq \varphi^{-1}(J)$. By the inclusion map, we have

$$\varphi(y) = \varphi(rx) = rx \in r\varphi^{-1}(J) \subseteq \varphi^{-1}(J) \implies y = \varphi(y) \in J.$$

Since $x \in R$ and $x \in J$, it follows that $x \in J \cap R$. So, we have $y \in J \cap R$ and so $r(J \cap R) \subseteq J \cap R$.

(ii) Let $y \in (J \cap R)r$. Then $y = zr$ for some $z \in J \cap R$. Now, $z \in J \cap R$ implies that $z \in J$ and $z \in R$. Note that by definition of the inclusion map,

$$\varphi(z) = z \in J \implies z \in \varphi^{-1}(J).$$

Since $\varphi^{-1}(J)$ is an ideal, it follows that $\varphi^{-1}(J)r \subseteq \varphi^{-1}(J)$. Thus, $y \in \varphi^{-1}(J)r \subseteq \varphi^{-1}(J)$. Thus, $y \in \varphi^{-1}(J)$ which implies that $\varphi(y) \in J$ and so $y \in J$ (by definition of the inclusion map). Since $z \in J \cap R$, it follows that $y \in R$. Thus, $y \in J \cap R$ and so $(J \cap R)r \subseteq J \cap R$.

Now, (i) and (ii) imply that $J \cap R$ is an ideal of R . ■

- (b) Prove that if φ is surjective and I is an ideal of R , then $\varphi(I)$ is an ideal of S . Give an example where this fails if φ is not surjective.

Proof. Since I is an ideal of R , we know that, since φ is a surjective homomorphism, $\varphi(I)$ is also a additive subgroup of S . Now, we will show that $\varphi(I)$ is closed under multiplication. Our goal is to show that for all $s \in S$,

$$(i) \quad s\varphi(I) \subseteq \varphi(I),$$

$$(ii) \quad \varphi(I)s \subseteq \varphi(I).$$

Let $s \in S$ be arbitrary. Let $y \in s\varphi(I)$. Then $y = sv$ for some $v \in \varphi(I)$. Since $v \in \varphi(I)$, there exists a $u \in I$ such that $v = \varphi(u)$. Since $s \in S$ and φ is surjective, it follows that there exists a $t \in R$ such that $\varphi(t) = s$. Then

$$y = sv = \varphi(s)\varphi(u) = \varphi(tu) \implies y = \varphi(tu).$$

Since I is an ideal, it follows that $tI \subseteq I$. Thus, $tu \in tI$ implies that $tu \in I$. This tells us that $y \in \varphi(I)$ and so $s\varphi(I) \subseteq \varphi(I)$.

Now, let $y \in \varphi(I)s$. Then $y = ws$ for some $w \in \varphi(I)$. Since $w \in \varphi(I)$, there exists a $z \in I$ such that $\varphi(z) = w$. Since $s \in S$ and φ is surjective, there exists $t \in R$ such that $\varphi(t) = s$. Thus,

$$y = ws = \varphi(z)\varphi(t) = \varphi(zs).$$

Now, we claim that $zs \in I$. Notice that $zs \in It$. Since I is an ideal, we have $It \subseteq I$, implying that $zs \in I$. Hence, $y \in \varphi(I)$ and so $\varphi(I)s \subseteq \varphi(I)$.

Now, (i) and (ii) imply that $\varphi(I)$ is an ideal in S . ■

Problem 5 (Exercise 26 from 7.3). The **characteristic** of a ring R is the smallest positive integer n such that $1 + 1 + \cdots + 1 = 0$ (n times) in R ; if no such integer exists the characteristic of R is said to be 0.

(a) Given that the map $\mathbb{Z} \rightarrow R$ defined by

$$k \mapsto \begin{cases} 1 + 1 + \cdots + 1 \text{ (} k \text{ times)} & \text{if } k > 0 \\ 0 & \text{if } k = 0 \\ -1 - 1 - 1 \cdots - 1 \text{ (} k \text{ times)} & \text{if } k < 0 \end{cases}$$

is a ring homomorphism, prove that its kernel is $n\mathbb{Z}$, where n is the characteristic of R .

Proof. Notice that

$$\begin{aligned} \ker(\varphi) &= \{k \in \mathbb{Z} : \varphi(k) = 0\} \\ &= \{k \in \mathbb{Z} : \underbrace{1 + 1 + \cdots + 1}_{k \text{ times}} = 0\} \\ &= \{k \in \mathbb{Z} : k = 0\} \\ &= \{k \in \mathbb{Z} : nk = 0\} && (\text{in } \mathbb{Z}/n\mathbb{Z}) \\ &= \{nk : k \in \mathbb{Z}\} \\ &= n\mathbb{Z}. \end{aligned}$$

(b) Determine the characteristics of the rings $\mathbb{Q}, \mathbb{Z}[x], \mathbb{Z}/n\mathbb{Z}[x]$.

Solution. $\mathbb{Q}$ is characteristic zero, $\mathbb{Z}[x]$ is characteristic zero since $\mathbb{Z}$ is characteristic zero, and $\mathbb{Z}/n\mathbb{Z}$ is characteristic n . ■

(c) Prove that if p is a prime and if R is a commutative ring of characteristic p , then $(a+b)^p = a^p + b^p$ for all $a, b \in R$.

Proof. By the Binomial Theorem and the fact that p is the characteristic of R , it follows that

$$\begin{aligned} (a+b)^p &= \sum_{k=0}^p \binom{p}{k} a^k b^{p-k} \\ &= \binom{p}{0} a^0 b^p + \binom{p}{1} a b^{p-1} + \binom{p}{2} a^2 b^{p-2} + \cdots + \binom{p}{p-1} a^{p-1} b^1 + \binom{p}{p} a^p b^0 \\ &= b^p + \underbrace{(pa)b^{p-1} + \frac{p(p-1)}{2} a^2 b^{p-2} + \cdots + pa^{p-1} b^1}_{0} + a^p && (R \text{ has characteristic } p) \\ &= b^p + a^p \\ &= a^p + b^p && (\text{addition is commutative}) \end{aligned}$$

Thus, we conclude that

$$(a+b)^p = a^p + b^p.$$

